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Load-bearing accessory, and window

Abstract

A load-bearing accessory includes a hinge, a frame and a connector. The frame includes an adjustment portion and a support portion. The distance between the adjustment portion and the support portion is adjustable. A first end of the connector is connected to the hinge, and a second end of the connector is connected to the adjustment portion. One of the hinge or the frame that is located at the upper position of the load-bearing accessory is configured to support the connector, and the connector is configured to support the other one of the hinge or the frame that is located at the lower position of the load-bearing accessory.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage filing, filed under 35 U.S.C. 371, of International Patent Application No. PCT/CN2023/078456 filed on Feb. 27, 2023, which claims priority to Chinese Patent Application No. 202210224162.8 filed with the China National Intellectual Property Administration (CNIPA) on Mar. 9, 2022, the disclosures of which is are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present application relates to the technical field of doors and windows, for example, a load-bearing accessory and a window.

BACKGROUND

(3) A door and window sash member is generally connected to a frame body through an open hinge or a hidden hinge. For structures connected through an open hinge, when a door and window sash is relatively heavy, only the number of hinges needs to be increased. For structures connected through a hidden hinge, due to the inherent structure of the hidden hinge, a door and window is usually provided with only a pair of hidden hinges. This limits the load-bearing capacity of the pair of hidden hinges and makes it difficult to achieve the load-bearing capacity of the open hinge.

(4) The load-bearing force of a hinge is shared through some additional load-bearing accessories. A load-bearing accessory uses a rigid shaft to connect a frame body and a sash member. The rigid shaft is relatively long, making the entire load-bearing accessory relatively long and relatively expensive, and a pure hinge structure can only be applied to a casement window and cannot be applied to a tilt and turn window.

SUMMARY

(5) The present application provides a load-bearing accessory and a window to solve the technical problems of high cost and inability to be applied to a tilt and turn window in the related art. A load-bearing accessory is provided and includes a hinge, a frame including an adjustment portion and a support portion, where a distance between the adjustment portion and the support portion is adjustable; and a connector, where a first end of the connector is connected to the hinge, a second end of the connector is connected to the adjustment portion, one of the hinge or the frame that is located at an upper position of the load-bearing accessory is configured to support the connector, and the connector is configured to support another one of the hinge or the frame that is located at a lower position of the load-bearing accessory.

(6) The frame includes an adjustment screw; a first threaded hole penetrating one of the adjustment portion or the support portion in an up-down direction is disposed inside the one of the adjustment portion or the support portion, the adjustment screw is threadedly connected to the first threaded hole and has an end portion abutting onto another one of the adjustment portion or the support portion, a first through hole is disposed inside the support portion, and the connector passes through the first through hole.

- (7) A second threaded hole is disposed inside the support portion, a periphery of the adjustment portion is provided with an external thread, and the adjustment portion is threadedly connected to the second threaded hole.
- (8) The adjustment portion includes an adjustment block and a spring, a stepped hole penetrating the adjustment block in an up-down direction is disposed inside the adjustment block, one end of the connector is located inside the stepped hole, the stepped hole includes a large diameter segment and a small diameter segment, the spring is located inside the large diameter segment, has a first end abutting onto a solid end surface of the small diameter segment in a peripheral direction of the small diameter segment and is sleeved outside the connector, and an end portion of the connector abuts onto a second end of the spring.
- (9) The one of the hinge or the frame that is located at the upper position of the load-bearing accessory is mounted on a first component, the another one of the hinge or the frame that is located at the lower position of the load-bearing accessory is mounted on a second component, and the first component is configured to support the second component.
- (10) The hinge includes a fixed portion and a connecting portion, the fixed portion is connected to the first component or the second component, the connecting portion includes a connecting block and a fixed shaft, the connecting block is provided with a placement groove, and a first connecting hole along a height direction, upper and lower sides of the placement groove are each provided with at least one first connecting hole, the fixed portion is provided with one second connecting hole along a height direction, the fixed shaft sequentially passes through the at least one first connecting hole on the upper side of the placement groove, the one second connecting hole, and the at least one first connecting hole on the lower side of the placement groove, the placement groove is able to accommodate the fixed portion, the connecting block is provided with a mounting hole penetrating the connecting block in an up-down direction, and the connector is connected to an inside of the mounting hole.
- (11) The fixed portion includes a first lock tab, a first fixed block and a first screw, the first lock tab is fixedly connected to one of the first component or the second component, the first fixed block is provided with a second through hole, and the first screw passes through the second through hole and is threadedly connected to the first lock tab.
- (12) The fixed portion includes a first fixed block and a first connecting screw, the first fixed block is provided with a first connecting through hole, and the first connecting screw passes through the first connecting through hole and is threadedly connected to one of the first component or the second component.
- (13) The support portion includes a second lock tab, a second fixed block and a second screw, the second lock tab is fixedly connected to another one of the first component or the second component, the second fixed block is provided with a third through hole, and the second screw passes through the third through hole and is threadedly connected to the second lock tab.
- (14) The support portion includes a second fixed block and a second connecting screw, the second fixed block is provided with a second connecting through hole, and the second connecting screw passes through the second connecting through hole and is threadedly connected to another one of the first component or the second component.
- (15) The connector is a flexible member.
- (16) A window is provided and includes a fixed window frame, a movable window sash and the preceding load-bearing accessory, where one of a hinge or a frame that is located at an upper position of the load-bearing accessory is connected to the fixed window frame, and another one of the hinge or the frame that is located at a lower position of the load-bearing accessory is connected to the movable window sash.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a view illustrating the structure of a window according to embodiment one of the present application.
- (2) FIG. 2 is a view illustrating the structure of a load-bearing accessory according to embodiment one of the present application.
- (3) FIG. 3 is an exploded view illustrating the structure of a load-bearing accessory according to embodiment one of the present application.
- (4) FIG. 4 is an exploded view illustrating the structure of another load-bearing accessory according to embodiment one of the present application.
- (5) FIG. 5 is section view one illustrating the partial structure of a load-bearing accessory according to embodiment one of the present application.
- (6) FIG. 6 is section view two illustrating the partial structure of a load-bearing accessory according to embodiment one of the present application.
- (7) FIG. 7 is a view illustrating the structure of a load-bearing accessory according to embodiment two of the present application.
- (8) FIG. 8 is a view illustrating the structure of a load-bearing accessory according to embodiment three of the present application.
- (9) FIG. 9 is a section view illustrating the partial structure of a load-bearing accessory according to embodiment three of the present application.
- (10) FIG. 10 is a view illustrating the partial structure of a load-bearing accessory according to embodiment three of the present application.
- (11) FIG. 11 is a view illustrating the structure of a load-bearing accessory according to embodiment four of the present application.

REFERENCE LIST

(12) **10** first component **101** fixed window frame **20** second component **201** movable window sash **1** hinge **11** fixed portion **111** first lock tab **112** first fixed block **113** first screw **114** second through hole **115** second connecting hole **116** first connecting screw **117** first connecting through hole **12** connecting portion **121** connecting block **1211** first connecting hole **1212** mounting hole **122** fixed shaft **123** placement groove **13** gasket **2** frame **21** adjustment portion **211** adjustment block **212** spring **213** adjustment groove **214** first threaded hole **215** stepped hole **216** external thread **22** support portion **221** second lock tab **222** second fixed block **223** second screw **224** third through hole **225** first through hole **226** second threaded hole **227** second connecting screw **228** second connecting through hole **23** adjustment screw **3** connector

DETAILED DESCRIPTION

(13) Examples of embodiments of the present application are described below in detail with reference to the drawings. The same or similar reference numerals indicate the same or similar elements or elements having the same or similar functions. The embodiments described below with reference to the drawings are exemplary and intended to explain the present application and cannot be construed as limiting the present application.

(14) In the description of the embodiments, terms “joined”, “connected”, and “fixed” are to be understood in a broad sense unless otherwise expressly specified and limited. For example, the term “connected” may refer to “fixedly connected”, “detachably connected”, or “integrated”, may refer to “mechanically connected” or “electrically connected”, or may refer to “connected directly”, “connected indirectly through an intermediary”, or “connected inside two elements” or “an interaction relation between two elements”. For those of ordinary skill in the art, meanings of the preceding terms in the present application may be understood based on situations.

(15) In the present application, unless otherwise specified and limited, when a first feature is described as being “on” or “below” a second feature, the first feature and the second feature may be in direct contact or be in contact via another feature between the two features instead of being in

direct contact. Moreover, when the first feature is described as being “on”, “above”, or “over” the second feature, the first feature is right on, above, or over the second feature, the first feature is obliquely on, above, or over the second feature, or the first feature is simply at a higher level than the second feature. When the first feature is described as being “under”, “below”, or “underneath” the second feature, the first feature is right under, below; or underneath the second feature, the first feature is obliquely under, below; or underneath the second feature, or the first feature is simply at a lower level than the second feature.

(16) In embodiments of the present application, a door and window structure or a door and window assembly is a structure or assembly applied to a door and window.

Embodiment One

(17) Referring to FIGS. 1 and 2, an embodiment of the present application provides a load-bearing accessory. The load-bearing accessory includes a hinge 1, a frame 2 and a connector 3. The frame 2 includes an adjustment portion 21 and a support portion 22. The distance between the adjustment portion 21 and the support portion 22 is adjustable. A first end of the connector 3 is connected to the hinge 1, and a second end of the connector 3 is connected to the adjustment portion 21. Located at the upper position of the load-bearing accessory, the frame 2 is configured to support the connector 3, and the connector 3 is configured to support the hinge 1. The frame 2 supports the connector 3 so that the connector 3 can support the hinge 1, and so that the frame 2 can bear the force of the hinge 1. Consequentially, the structure can be simple, and the cost can be low. Since the distance between the adjustment portion 21 and the support portion 22 is adjustable, the distance between the hinge 1 and the frame 2 is changeable, that is, the support may also be maintained when the distance between the hinge 1 and the frame 2 becomes longer.

(18) The connector 3 is resiliently connected to the adjustment portion 21 to cooperate with the distance adjustment between the adjustment portion 21 and the support portion 22 so that the adjustment portion 21 can adjust the magnitude of the force shared by the load-bearing accessory.

(19) Referring to FIGS. 2 to 4, the frame 2 includes an adjustment screw 23, a first threaded hole 214 penetrating the adjustment portion 21 in an up-down direction is disposed inside the adjustment portion 21, and the adjustment screw 23 is threadedly connected to the first threaded hole 214 and has an end portion abutting onto the support portion 22 so that the distance between the adjustment portion 21 and the support portion 22 can be adjusted. Moreover, a first through hole 225 is disposed inside the support portion 22, the connector 3 passes through the first through hole 225, and the first through hole 225 limits the position of the connector 3 so that the adjustment portion 21 cannot move with the support portion 22 in a horizontal direction.

(20) The adjustment portion 21 includes an adjustment block 211 and a spring 212. A stepped hole 215 penetrating the adjustment block 211 in an up-down direction is disposed inside the adjustment block 211. One end of the connector 3 is located inside the stepped hole 215. The stepped hole 215 includes a large diameter segment and a small diameter segment. The spring 212 is located inside the large diameter segment, has a first end abutting onto a solid end surface of the small diameter segment in a peripheral direction of the small diameter segment and is sleeved outside the connector 3. An end portion of the connector 3 abuts onto a second end of the spring 212. Consequentially, the connector 3 can be resiliently connected to the adjustment portion 21 so that the adjustment portion 21 can adjust the magnitude of the force shared by the load-bearing accessory; the materials of the spring 212 can be conveniently obtained, and the cost can be low; and the stepped hole 215 can limit the position of the spring 212 so that the spring 212 can be placed inside the adjustment block 211.

(21) The adjustment screw 23 is rotated so that the adjustment block 211 can move away from the support portion 22. The end portion of the connector 3 compresses the spring 212 to tighten the connection between the connector 3 and the adjustment portion 21.

(22) A head of the adjustment screw 23 is provided with an adjustment groove, and the adjustment screw 23 is rotated through a screwdriver adapted to the shape and size of the adjustment groove.

In this embodiment, the adjustment groove is in a hexagon shape. Optionally, the adjustment groove may be in other shapes such as a minus or a square.

(23) Referring to FIG. 1, one of the hinge **1** or the frame **2** that is located at the upper position of the load-bearing accessory is mounted on a first component **10**, and the other one of the hinge **1** or the frame **2** that is located at the lower position of the load-bearing accessory is mounted on a second component **20**, and the first component **10** is configured to support the second component **20**. In this embodiment, the first component **10** is a fixed window frame, and the second component **20** is a movable window sash. Optionally, in other embodiments, the first component **10** and the second component **20** may also be other structures. For example, the first component **10** is a door frame, and the second component **20** may be a door body.

(24) Referring to FIGS. 3 and 5, the hinge **1** includes a fixed portion **11** and a connecting portion **12**. The fixed portion **11** is connected to the second component **20**. The connecting portion **12** includes a connecting block **121** and a fixed shaft **122**. The connecting block **121** is provided with a placement groove **123**, and a first connecting hole **1211** along a height direction. Upper and lower sides of the placement groove **123** are each provided with at least one first connecting hole **1211**. The fixed portion **11** is provided with one second connecting hole **115** along a height direction. The fixed shaft **122** sequentially passes through the at least one first connecting hole **1211** on the upper side of the placement groove **123**, the second connecting hole **115**, and the at least one first connecting hole **1211** on the lower side of the placement groove **123**. The placement groove **123** can accommodate the fixed portion **11**. The connecting block **121** is provided with a mounting hole **1212** penetrating the connecting block **121** in an up-down direction. The connector **3** is connected inside of the mounting hole **1212**. Consequentially, the fixed shaft **122** enables the rotation between the connecting block **121** and the fixed portion **11**. In this embodiment, the placement groove **123** is a U-shaped groove, and the mounting hole **1212** is a stepped hole preventing the mounting hole **1212** the connector **3** from disengaging from the connecting block **121**. The movable window sash is provided with a sealing strip. The surface of the connecting block **121** facing the sealing strip is curved, preventing wear on the sealing strip when the movable window sash is opened and closed.

(25) Referring to FIGS. 3 and 4, the fixed portion **11** includes a first lock tab **111**, a first fixed block **112** and a first screw **113**. The first lock tab **111** engages with the second component **20**, facilitating disassembly and assembly. The first fixed block **112** is provided with a second through hole **114**. The first screw **113** passes through the second through hole **114** and is threadedly connected to the first lock tab **111**. Consequentially, the structure can be simple, and convenient to mount, and the hinge **1** can be connected to the second component **20**. Two first screws **113** are provided along the height direction of the first lock tab **111**. The number of first screws **113** may be selected according to actual situations.

(26) Optionally, the fixed portion **11** includes the first fixed block **112** and a first connecting screw **116**. The first fixed block **112** is provided with a first connecting through hole **117**. The first connecting screw **116** passes through the first connecting through hole **117** and is threadedly connected to the second component **20**. That is, the first fixed block **112** can also be connected to the second component **20** without the first lock tab **111**.

(27) The hinge **1** further includes two gaskets **13**. The two gaskets **13** are placed inside the placement groove **123** and are located on the upper and lower sides of the first fixed block **112** respectively. The fixed shaft **122** passes through the two gaskets **13** to prevent wear at the rotation junctures between the first fixed block **12** and the connecting block **121**.

(28) The support portion **22** includes a second lock tab **221**, a second fixed block **222** and a second screw **223**. The second lock tab **221** engages with the first component **10**. The second fixed block **222** is provided with a third through hole **224**. The second screw **223** passes through the third through hole **224** and is threadedly connected to the second lock tab **221**. Consequentially, the structure can be simple, and convenient to mount, and the frame **2** can be connected to the first component **10**. Two second screws **223** are provided along the height direction of the second lock

tab **221**. The number of second screws **223** may be selected according to actual situations.

(29) Optionally, the support portion **22** includes the second fixed block **222** and a second connecting screw **227**. The second fixed block **222** is provided with a second connecting through hole **228**. The second connecting screw **227** passes through the second connecting through hole **228** and is threadedly connected to the first component **10**. That is, the second fixed block **222** may also be connected to the first component **10** without the second lock tab **221**.

(30) Referring to FIGS. **2** to **5**, the connector **3** is a flexible member. The flexible member is not prone to damage when being pulled so that the service life of the connector **3** can be prolonged. The connector **3** includes a body and two heads. The two heads are connected to two ends of the body respectively. The body is a steel cable. The two heads are stainless steel members. The cross sections of the body and the two heads are circular. The diameter of each head is greater than the diameter of the body. The two heads are located inside the stepped hole **215** and the mounting hole **1212** respectively. During assembly, the body first passes through the stepped hole **215** and the mounting hole **1212**, the spring **212** is sleeved outside the body, and the two heads are then mounted at the two ends of the body respectively.

(31) An embodiment of the present application provides a window. The window includes a fixed window frame **101**, a movable window sash **201** and the load-bearing accessory. The frame **2** is connected to the fixed window frame **101**. The hinge **1** is connected to the movable window sash **201**.

(32) The fixed window frame **101** can support the movable window sash **201** through the load-bearing accessory. When the tilt and turn window is opened, the distance between the fixed window frame **101** and the movable window sash **201** is changed. Since the distance between the adjustment portion **21** and the support portion **22** is adjustable, the support can also be maintained even when the distance between the hinge **1** and the frame **2** becomes longer. That is, the load-bearing accessory can be applied to both the casement window and the tilt and turn window and thereby have a wide application range. The connector **3** is resiliently connected to the adjustment portion **21** and is a flexible member so that the load-bearing accessory can be easily opened when being applied to the tilt and turn window. The fixed window frame **101** is connected to the movable window sash **201** through a hidden hinge, and the load-bearing accessory shares the load-bearing force of the hidden hinge so that the fixed window frame **101** can bear the movable window sash **201** having a large weight. The service life of the hidden hinge is prolonged so that the hidden hinge cannot be prone to damage.

Embodiment Two

(33) FIG. **6** shows embodiment two in which the same or corresponding parts as in embodiment one use reference numerals corresponding to embodiment one. For simplicity; only differences between embodiment two and embodiment one are described. The differences are that the first threaded hole **214** penetrating the support portion **22** in the up-down direction is disposed inside the support portion **22**, and the adjustment screw **23** is threadedly connected to the first threaded hole **214** and has an end portion abutting onto the adjustment portion **21**. The first threaded hole **214** is disposed inside the support portion **22** so that the distance between the adjustment portion **21** and the support portion **22** can also be changed, thereby tightening the connection between the connector **3** and the adjustment portion **21**.

Embodiment Three

(34) FIGS. **7** to **9** show embodiment three in which the same or corresponding parts as in embodiment one use reference numerals corresponding to embodiment one. For simplicity, only differences between embodiment three and embodiment one are described. The differences are that a second threaded hole **226** is disposed inside the support portion **22**, a periphery of the adjustment portion **21** is provided with an external thread **216**, and the adjustment portion **21** is threadedly connected to the second threaded hole **226**. The external thread **216** is disposed on the adjustment block **211**, and the adjustment block **211** can move away from the support portion **22** when the

adjustment block **211** is rotated. The end portion of the connector **3** compresses the spring **212** to tighten the connection between the connector **3** and the adjustment portion **21**.

(35) One end of the adjustment block **211** facing away from the hinge **1** is provided with an adjustment groove **213**. The adjustment groove **213** communicates with the stepped hole **215**. The adjustment block **211** is rotated through a screwdriver adapted to the shape and size of the adjustment groove **213**. In this embodiment, the adjustment groove **213** is in a hexagon shape. Optionally, the adjustment groove **213** may be in other shapes such as a minus, a cross and a square, as long as the lower part of the adjustment groove **213** is ensured to have an entity to support the screwdriver to perform the adjustment.

Embodiment Four

(36) FIG. **10** shows embodiment four in which the same or corresponding parts as in embodiment one use reference numerals corresponding to embodiment one. For simplicity, only differences between embodiment four and embodiment one are described. The differences are that the hinge **1** is located above the frame **2** and is connected to the first component **10**, and the frame **2** is connected to the second component **20**. The first lock tab **111** is fixedly connected to the first component **10**, and the second lock tab **221** is fixedly connected to the second component **20**, that is, the hinge **1** is connected to the fixed window frame **101**, and the frame **2** is connected to the movable window sash **201**. As long as rotation is ensured between the first fixed block **112** and the connecting block **121** of the hinge **1** when the movable window sash **201** is opened, the position of the connecting block **121** can remain unchanged, the connecting block **121** can prevent the sealing strip on the movable window sash **201** from being worn by the connector **3** during the opening and closing of the movable window sash **201**, and the hinge **1** can protect the sealing strip on the movable window sash **201**.

(37) The present application provides the load-bearing accessory. The one of the hinge or the frame that is located at the upper position of the load-bearing accessory supports the connector so that the connector can support the other one of the hinge or the frame that is located at the lower position of the load-bearing accessory, and so that the one of the hinge or the frame that is located at the upper position of the load-bearing accessory can bear the force of the other one of the hinge or the frame that is located at the lower position of the load-bearing accessory. Consequentially, the structure can be simple, and the cost can be low. The distance between the adjustment portion and the support portion is adjustable so as to adjust two ends of the connector to suitable positions for support. The distance between the hinge and the frame is changeable, that is, the support can also be maintained even when the distance between the hinge and the frame becomes longer.

Claims

1. A load-bearing accessory, comprising: a hinge, wherein the hinge comprises a fixed portion and a connecting portion, the connecting portion comprises a connecting block that is rotatably connected to a fixed shaft; a frame comprising an adjustment portion and a support portion, wherein a distance between the adjustment portion and the support portion is adjustable; wherein the fixed portion is connected to a first component, and the frame is connected to a second component; or the fixed portion is connected to the second component, and the frame is connected to the first component; and a connector, wherein a first end of the connector is connected to the connecting block, and a second end of the connector is connected to the adjustment portion; and one of the hinge or the frame that is located at an upper position of the load-bearing accessory is configured to support the connector, and the connector is configured to support another one of the hinge or the frame that is located at a lower position of the load-bearing accessory; wherein the connecting block is provided with a placement groove, and first connecting holes along a up-down direction, upper and lower sides of the placement groove are each provided with at least one first connecting hole, the fixed portion is provided with one second connecting hole along the up-down direction,

the fixed shaft sequentially passes through one of the at least one first connecting hole on the upper side of the placement groove, the one second connecting hole, and one of the at least one first connecting hole on the lower side of the placement groove, the placement groove is able to accommodate the fixed portion, the connecting block is provided with a mounting hole penetrating the connecting block in the up-down direction, and the connector is connected to an inside of the mounting hole.

2. The load-bearing accessory according to claim 1, wherein the frame comprises an adjustment screw, a first threaded hole penetrating one of the adjustment portion or the support portion in the up-down direction is disposed inside the one of the adjustment portion or the support portion, the adjustment screw is threadedly connected to the first threaded hole and has an end portion abutting onto another one of the adjustment portion or the support portion, a first through hole is disposed inside the support portion, and the connector passes through the first through hole.

3. The load-bearing accessory according to claim 1, wherein a second threaded hole is disposed inside the support portion, a periphery of the adjustment portion is provided with an external thread, and the adjustment portion is threadedly connected to the second threaded hole.

4. The load-bearing accessory according to claim 2, wherein the adjustment portion comprises an adjustment block and a spring, a stepped hole penetrating the adjustment block in the up-down direction is disposed inside the adjustment block, one end of the connector is located inside the stepped hole, the stepped hole comprises a large diameter segment and a small diameter segment, the spring is located inside the large diameter segment, has a first end abutting onto a solid end surface of the small diameter segment in a peripheral direction of the small diameter segment, and is sleeved outside the connector, and an end portion of the connector abuts onto a second end of the spring.

5. The load-bearing accessory according to claim 1, wherein the fixed portion comprises a first lock tab, a first fixed block and a first screw, the first lock tab engages with one of the first component or the second component, the first fixed block is provided with a second through hole, and the first screw passes through the second through hole and is threadedly connected to the first lock tab.

6. The load-bearing accessory according to claim 1, wherein the fixed portion comprises a first fixed block and a first connecting screw, the first fixed block is provided with a first connecting through hole, and the first connecting screw passes through the first connecting through hole and is threadedly connected to one of the first component or the second component.

7. The load-bearing accessory according to claim 5, wherein the support portion comprises a second lock tab, a second fixed block and a second screw, the second lock tab engages with another one of the first component or the second component, the second fixed block is provided with a third through hole, and the second screw passes through the third through hole and is threadedly connected to the second lock tab.

8. The load-bearing accessory according to claim 5, wherein the support portion comprises a second fixed block and a second connecting screw, the second fixed block is provided with a second connecting through hole, and the second connecting screw passes through the second connecting through hole and is threadedly connected to another one of the first component or the second component.

9. The load-bearing accessory according to claim 1, wherein the connector is a flexible member.

10. A window, comprising a fixed window frame, a movable window sash, and the load-bearing accessory according to claim 1, wherein one of the hinge or the frame that is located at the upper position of the load-bearing accessory is connected to the fixed window frame, and another one of the hinge or the frame that is located at the lower position of the load-bearing accessory is connected to the movable window sash.

11. The load-bearing accessory according to claim 3, wherein the adjustment portion comprises an adjustment block and a spring, a stepped hole penetrating the adjustment block in the up-down direction is disposed inside the adjustment block, one end of the connector is located inside the

stepped hole, the stepped hole comprises a large diameter segment and a small diameter segment, the spring is located inside the large diameter segment, has a first end abutting onto a solid end surface of the small diameter segment in a peripheral direction of the small diameter segment, and is sleeved outside the connector, and an end portion of the connector abuts onto a second end of the spring.

12. The load-bearing accessory according to claim 6, wherein the support portion comprises a second lock tab, a second fixed block and a second screw, the second lock tab engages with another one of the first component or the second component, the second fixed block is provided with a third through hole, and the second screw passes through the third through hole and is threadedly connected to the second lock tab.

13. The load-bearing accessory according to claim 6, wherein the support portion comprises a second fixed block and a second connecting screw, the second fixed block is provided with a second connecting through hole, and the second connecting screw passes through the second connecting through hole and is threadedly connected to another one of the first component or the second component.

14. The window according to claim 10, wherein the frame comprises an adjustment screw, a first threaded hole penetrating one of the adjustment portion or the support portion in the up-down direction is disposed inside the one of the adjustment portion or the support portion, the adjustment screw is threadedly connected to the first threaded hole and has an end portion abutting onto another one of the adjustment portion or the support portion, a first through hole is disposed inside the support portion, and the connector passes through the first through hole.

15. The window according to claim 10, wherein a second threaded hole is disposed inside the support portion, a periphery of the adjustment portion is provided with an external thread, and the adjustment portion is threadedly connected to the second threaded hole.

16. The window according to claim 14, wherein the adjustment portion comprises an adjustment block and a spring, a stepped hole penetrating the adjustment block in the up-down direction is disposed inside the adjustment block, one end of the connector is located inside the stepped hole, the stepped hole comprises a large diameter segment and a small diameter segment, the spring is located inside the large diameter segment, has a first end abutting onto a solid end surface of the small diameter segment in a peripheral direction of the small diameter segment, and is sleeved outside the connector, and an end portion of the connector abuts onto a second end of the spring.
